

ALGORITHM
Final Exam, June 24, 2021
5:30-8:30 pm

1. (15%) Please answer the following questions:
 - (a) Illustrate the operation of RADIX-SORT on the following numbers: 268, 356, 793, 254, 257, 399, 358, 921, 331, 536.
 - (b) Show that if a node in a binary search tree has two children, then its successor has no left child.
 - (c) For the set of keys $\{5, 22, 13, 15, 27, 33, 11\}$, draw binary search trees of height 2, 3, 4, 5 and 6.
2. (15%) Given a sequence $X = \langle x_1, x_2, \dots, x_m \rangle$, the longest increasing subsequence problem is to find an increasing subsequence of X with the longest length.
 - (a) Find a longest increasing subsequence of $\langle 6, 35, 24, 7, 8, 26, 21, 34 \rangle$.
 - (b) Describe an algorithm that solves the longest increasing subsequence problem in $O(n^2)$ time.
3. (20%) Consider inserting the keys 5, 22, 13, 15, 27, 33, 11, 21 into a hash table of length $m = 13$ using open addressing with the auxiliary hash function $h'(k) = k$. Show the results after the insertion of each number using the following methods.
 - (a) Linear hashing.
 - (b) Quadratic probing with $c_1 = 1$ and $c_2 = 3$.
 - (c) Double hashing with $h_1(k) = k$ and $h_2(k) = 1 + (k \bmod (m - 1))$.
 - (d) Chaining.
4. (15%) Medians and Order Statistics.
 - (a) In the algorithm SELECT, the input elements are divided into groups of 5. Will the algorithm work in linear time if they are divided into groups of 9?
 - (b) Show how QUICKSORT can be made to run in $O(n \lg n)$ time in the worst case.
5. (15%) Red-Black Trees.
 - (a) Show the red-black trees that result after successively inserting the keys 5, 22, 13, 15, 27, 33, 11, 21 into an initially empty red-black tree.
 - (b) What is the largest possible number of internal nodes in a red-black tree with black height k ? What is the smallest possible number?

6. (15%) Minimizing lateness problem.

- Single resource processes one job at a time.
- Job i requires t_i units of processing time and is due at time d_i .
- If i starts at time s_i , it finishes at time $f_i = s_i + t_i$.
- Lateness: $\ell_i = \max\{0, f_i - d_i\}$.
- Goal: schedule all jobs to minimize maximum lateness $L = \max\{\ell_i \mid 1 \leq i \leq n\}$.

i	1	2	3	4	5	6	7
t_i	3	7	4	1	2	7	5
d_i	5	9	8	8	9	13	11

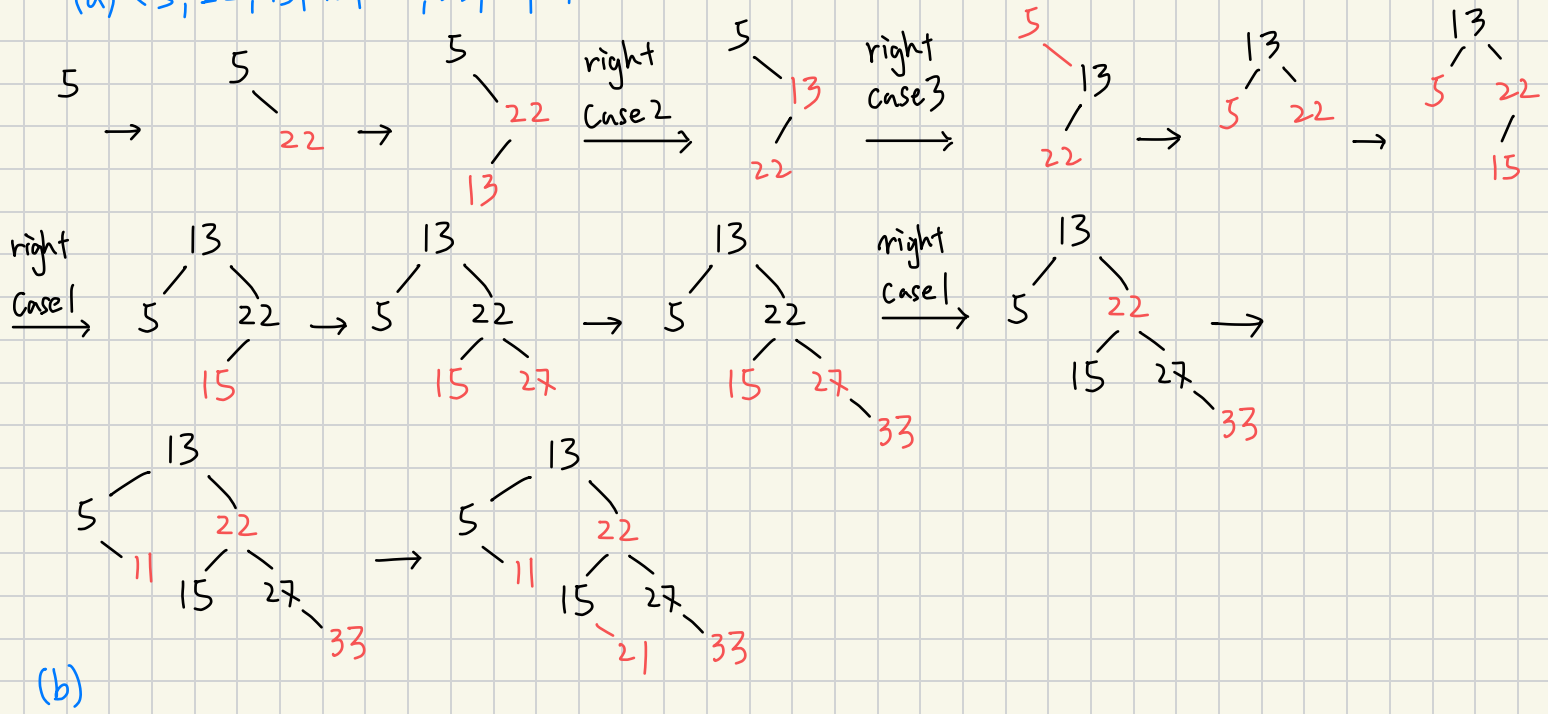
7. (20%) Alice is playing a game where she controls a character to walk on a $m \times n$ table. The character starts at the bottom-left corner (coordinate $(1, 1)$) and wants to go to the top-right corner (coordinate (m, n)). The character can only move to the right (from (i, j) to $(i, j + 1)$) or up (from (i, j) to $(i + 1, j)$). Each square of the table has a non-negative value $v[i, j]$. The goal is to find a path from the starting point to the end point that maximizes the sum of $v[i, j]$ for all the squares (i, j) on the path. See the left figure below, the arrows show the optimal path with the maximum total sum of all its values.

- Define the subproblems, write the recursive formula, and specify the base cases.
- Design an algorithm that outputs the maximum possible value. Your algorithm should run in time $O(mn)$.
- Suppose the character learned two new moves that allows it to move from (i, j) to $(i + 2, j - 1)$ and $(i - 1, j + 2)$ (See the right figure below). Explain how to modify the recursive formula and the algorithm.

2	0	0	3
10	2	2	1
1	3	20	0
1	2	0	3

2	0	0	3
10	2	2	1
1	3	20	0
1	2	0	3

5. (a) $\langle 5, 22, 13, 15, 27, 33, 11, 21 \rangle$



(b)

smallest possible number of internal nodes: $2^k - 1$
 largest possible number of internal nodes: $2^{2k} - 1$